

The number of partitions of the hypercube into squares: the asymptotic conjectured by Alon, Balogh and Potapov

github.com/graph-theory-AI

Machine-generated note

9 September 2026

Abstract

Let $f_2(d)$ be the number of partitions of the vertex set of the hypercube Q_d into 2-dimensional subcubes. Alon, Balogh and Potapov asked for the asymptotic behaviour of $f_2(d)$ and conjectured that it obeys the same bound as their Proposition 1.10 for partitions into singletons and squares, namely $f_2(d) = N^{1 \pm o(1)}$ with $N = (d/e)^{2^{d-1}}$, equivalently $\log f_2(d) = (1 + o(1)) 2^{d-1} \log d$. We prove this. The upper bound is their Proposition 1.6. For the lower bound, a pigeonhole argument selects a common singleton set L for many singleton-and-square partitions of Q_{d-2} ; placing four such partitions on the four layers of $Q_{d-2} \times Q_2$ and replacing the four aligned copies of each singleton by one vertical square yields $f_2(d) \geq (f_{0,2}(d-2)/2^{2^{d-2}})^4$, and the pigeonhole loss is negligible against the main term.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 4](#) records the full provenance.

1 Introduction

Let $Q_d = \{0, 1\}^d$ be the d -dimensional hypercube. A (k -dimensional) *subcube* of Q_d is a set of the form $\{x \in Q_d : x_i = c_i \text{ for } i \notin F\}$ for a set $F \subseteq [d]$ of k free coordinates and fixed values $c_i \in \{0, 1\}$; for $k = 2$ we also say *square*. Following Alon, Balogh and Potapov [1], for $S \subseteq \{0, 1, \dots, d\}$ let $f_S(d)$ be the number of partitions of Q_d into subcubes whose dimensions all lie in S ; we write $f_2(d) = f_{\{2\}}(d)$ and $f_{0,2}(d) = f_{\{0,2\}}(d)$, so that f_2 counts partitions into squares and $f_{0,2}$ counts partitions into singletons and squares. As in [1], put

$$n = 2^{d-1}, \quad N = N_d = \left(\frac{d}{e}\right)^n.$$

All logarithms are in a fixed base.

Two results of [1] are used. Proposition 1.6 there gives, for the total number $f(d)$ of partitions into subcubes of any dimensions,

$$f_2(d) \leq f(d) \leq (d+1)^{2^{d-1}}, \quad (1)$$

and Proposition 1.10 there (Proposition 1.7 with $r = 2$ in arXiv version 3, see [Remark 3.3](#)) states $f_{0,2}(m) = N_m^{1+o(1)}$; its proof gives in particular

$$\log f_{0,2}(m) \geq (1 - o(1)) 2^{m-1} \log m. \quad (2)$$

After these results the authors write: “It is natural to ask what happens if only 2-dimensional subcubes are allowed to be in a partition. We believe that the same bound as in Proposition [1.10] holds.”

Problem 1.1 ([1, Problem 1.11 in arXiv v1/v2; Problem 1.9 in the published version]). Determine the asymptotic behaviour of the number $f_2(d)$.

We prove the conjectured asymptotic.

Theorem 1.2 (Main theorem).

$$\log f_2(d) = (1 + o(1)) 2^{d-1} \log d \quad (d \rightarrow \infty),$$

equivalently $f_2(d) = d^{(1+o(1))2^{d-1}}$.

Remark 1.3 (Equivalence with the source paper's formulation; from the referee's report). Since $N^{1 \pm o(1)} = \exp((1 \pm o(1)) n(\log d - \log e))$ and $\log e / \log d \rightarrow 0$, the statement $f_2(d) = d^{(1 \pm o(1))n}$ is identical to $f_2(d) = N^{1 \pm o(1)}$: the constant-per-vertex factor e^{-n} is absorbed by the $o(1)$ in the exponent in both directions. Thus [Theorem 1.2](#) is exactly the bound of Proposition 1.10 of [1] transferred to f_2 , at the precision at which that proposition is stated. Finer, second-order factors of the form $c^{2^{d-1}}$ are not determined by [Theorem 1.2](#); see [Remark 3.2](#).

Idea of proof. The upper bound is (1). For the lower bound, set $m = d - 2$. Among the $f_{0,2}(m)$ singleton-and-square partitions of Q_m , at most 2^{2^m} different singleton sets $L \subseteq Q_m$ occur, so some L is the singleton set of at least $f_{0,2}(m)/2^{2^m}$ of them ([Lemma 2.1](#)). Identify $Q_{m+2} = Q_m \times Q_2$, put one such partition on each of the four layers $Q_m \times \{z\}$, keep its squares, and replace the four aligned singletons (x, z) , $z \in Q_2$, for each $x \in L$, by the vertical square $\{x\} \times Q_2$. The result is a square partition of Q_{m+2} , and the four layer partitions can be read off from it, so $f_2(m+2) \geq (f_{0,2}(m)/2^{2^m})^4$. Taking logarithms and using (2), the pigeonhole loss $2^d \log 2$ is $O(2^d)$ while the main term is of order $2^d \log d$. [\[full proof\]](#)

2 The lifting lemma

For a partition $\mathcal{P}$ of Q_m into singletons and squares, let

$$L(\mathcal{P}) = \{x \in Q_m : \{x\} \text{ is a part of } \mathcal{P}\}$$

be its *singleton set*.

Lemma 2.1 (Lifting lemma). *For every $m \geq 1$,*

$$f_2(m+2) \geq \left(\frac{f_{0,2}(m)}{2^{2^m}} \right)^4.$$

Idea of proof. Pigeonhole on singleton sets gives $L \subseteq Q_m$ with $a_L \geq f_{0,2}(m)/2^{2^m}$ partitions having $L(\mathcal{P}) = L$. Any ordered 4-tuple $(\mathcal{P}_z)_{z \in Q_2}$ of them is lifted to the square partition of $Q_m \times Q_2$ consisting of the squares $C \times \{z\}$ for C a square of $\mathcal{P}_z$ and the vertical squares $\{x\} \times Q_2$ for $x \in L$. The vertical squares are exactly the parts with free coordinates $m+1, m+2$, so L is recovered, and the horizontal squares in layer z recover $\mathcal{P}_z$; hence the lifting is injective and $f_2(m+2) \geq a_L^4$. [\[full proof\]](#)

3 Remarks

Remark 3.1 (An elementary upper bound). The writeup also records a self-contained upper bound that suffices for [Theorem 1.2](#), so that only the lower bound genuinely depends on [1]. Put $n = 2^{d-1}$, the number of even-weight vertices. Every square of Q_d contains exactly two even-weight vertices (antipodal in the square), so a square partition has $2^d/4 = n/2$ squares, and each square is determined by its smaller even-weight vertex (in a fixed ordering) together with its pair of free coordinates. This is an injection of the set of square partitions into a set of size at most

$$\binom{n}{n/2} \binom{d}{2}^{n/2} \leq 2^n d^n = \exp(n \log d + O(n)),$$

which gives $\limsup_{d \rightarrow \infty} \log f_2(d) / (2^{d-1} \log d) \leq 1$ directly.

Remark 3.2 (What remains open). [Theorem 1.2](#) determines the leading term $2^{d-1} \log d$ of $\log f_2(d)$. It does not determine a possible second-order factor of the form $c^{2^{d-1}}$; [Lemma 2.1](#) loses exactly such a factor (2^{-2^d}) through the pigeonhole step, and (2) is itself only stated up to $N_m^{o(1)}$. The referee notes that the authors of [1] were also interested in this finer question, which under a maximalist reading of “determine the asymptotic behaviour” stays open.

Remark 3.3 (Numbering of the cited propositions; from the referee’s report). The referee compared the TeX sources of arXiv versions 1, 2 and 3 of [1]. The lower bound (2) is Proposition 1.10 (environment zero2) in v1 and v2, with proof $f_{0,2}(m) \geq \binom{m}{2}^{(1/2-\delta)2^{m-1}}$ for every fixed $\delta > 0$; in v3 it is subsumed into Proposition 1.7, stated for every fixed $r \geq 2$, whose case $r = 2$ is the identical bound. The upper bound $f(d) \leq (d+1)^n$ is Proposition 1.6 in v2 and v3 (in v1 it is Theorem 1.6 with the weaker bound $(5e)^n N$, still sufficient here). The proof of (2) in [1] rests on a nibble-type counting theorem of Asratian and Kuzjurin [2], which neither the writeup nor the referee re-proved.

Remark 3.4 (Computational checks; from the referee’s report). Exhaustive exact-cover enumeration gave $f_{0,2}(1), \dots, f_{0,2}(4) = 1, 2, 10, 339$ and $f_2(2), \dots, f_2(5) = 1, 3, 30, 4380$; every enumerated square partition of Q_d has $2^d/4$ parts and $f_2(d) \leq (d+1)^{2^{d-1}}$ holds at every computed value. For $m = 3$ the best singleton set is $L = \emptyset$ with $a_L = 3$, and all $3^4 = 81$ lifted objects are valid, pairwise distinct square partitions of Q_5 appearing in the census; for $m = 4$ the best L is again $\emptyset$ with $a_L = 30$, and all $30^4 = 810,000$ lifted objects are valid, pairwise distinct square partitions of Q_6 , so $f_2(6) \geq 810,000$. At these sizes the right-hand side of [Lemma 2.1](#) is below 1, as expected: the pigeonhole cost only becomes negligible asymptotically.

Remark 3.5 (Novelty). The referee’s search ([Appendix B](#)) found no publication of this observation and no resolution of [Problem 1.1](#) among the papers citing [1]; the nearest related works (partitions of $\mathbb{Z}_q^n$ into subcubes of dimension $n - m$; partitions of $\mathbb{F}_2^m$ into 2-dimensional affine subspaces) concern different families. This has not been checked by a human.

4 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The argument was found by the language model gpt-5.6-so1 in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2401.00299__02; original writeup in attacks/2401.00299__02/output.md). (2) An independent adversarial referee agent (claude-fable-5) re-derived every step, checked the two cited propositions verbatim against all three arXiv versions of the source paper, and machine-verified the construction: all $30^4 = 810,000$ lifted 4-tuples at $m = 4$ give distinct valid square partitions of Q_6 , and exhaustive censuses gave $f_2(2), \dots, f_2(5) = 1, 3, 30, 4380$. Its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised with full proofs in [Appendix A](#). No mathematical content was changed. Three clarifications taken from the referee’s report are added as remarks and marked as such: the equivalence of the writeup’s asymptotic with the source paper’s $N^{1 \pm o(1)}$ formulation ([Remark 1.3](#)), the version-dependence of the proposition numbering ([Remark 3.3](#)), and the exact numbers of the machine checks ([Remark 3.4](#)). **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

A Full proofs

Lemma 2.1 (Lifting lemma). *For every $m \geq 1$,*

$$f_2(m+2) \geq \left(\frac{f_{0,2}(m)}{2^{2^m}} \right)^4.$$

Proof. For $L \subseteq Q_m$ let a_L be the number of partitions $\mathcal{P}$ of Q_m into singletons and squares with $L(\mathcal{P}) = L$. There are at most 2^{2^m} subsets L of Q_m , and $\sum_{L \subseteq Q_m} a_L = f_{0,2}(m)$, so some L satisfies

$$a_L \geq \frac{f_{0,2}(m)}{2^{2^m}}. \quad (3)$$

Fix such an L and identify Q_{m+2} with $Q_m \times Q_2$, the last two coordinates forming the Q_2 factor. Let $(\mathcal{P}_z)_{z \in Q_2}$ be an ordered 4-tuple of partitions counted by a_L , so that $L(\mathcal{P}_z) = L$ for all four z . Define a family $\Phi(\mathcal{P}_z)_z$ of subsets of $Q_m \times Q_2$ as follows:

1. for every $z \in Q_2$ and every square C of $\mathcal{P}_z$, the set $C \times \{z\}$;
2. for every $x \in L$, the set $\{x\} \times Q_2$.

Each set of type 1 is a square of Q_{m+2} (its two free coordinates are those of C , in $[m]$), and each set of type 2 is a square of Q_{m+2} (free coordinates $m+1$ and $m+2$).

The family is a partition. Let $(x, z) \in Q_m \times Q_2$. If $x \notin L$, then x is not a singleton of $\mathcal{P}_z$, so it lies in a unique square C of $\mathcal{P}_z$, and (x, z) lies in $C \times \{z\}$; it lies in no other set of type 1 (the squares of $\mathcal{P}_z$ are disjoint, and sets $C' \times \{z'\}$ with $z' \neq z$ miss the layer z) and in no set of type 2 (as $x \notin L$). If $x \in L$, then (x, z) lies in $\{x\} \times Q_2$, in no other set of type 2, and in no set of type 1, because x is a singleton of $\mathcal{P}_z$ and hence in no square of $\mathcal{P}_z$. So every vertex lies in exactly one set of the family.

The map is injective. Given the partition $\Phi(\mathcal{P}_z)_z$, the parts whose free coordinates are $\{m+1, m+2\}$ are exactly the sets of type 2, which recover L . For each $z \in Q_2$, the parts contained in the layer $Q_m \times \{z\}$ are exactly the sets $C \times \{z\}$ with C a square of $\mathcal{P}_z$, which recover the squares of $\mathcal{P}_z$; together with the singletons $\{x\}$, $x \in L$, these are all parts of $\mathcal{P}_z$. Hence distinct 4-tuples give distinct square partitions of Q_{m+2} , and

$$f_2(m+2) \geq a_L^4.$$

Combining with (3) gives the lemma. $\square$

Theorem 1.2 (Main theorem).

$$\log f_2(d) = (1 + o(1)) 2^{d-1} \log d \quad (d \rightarrow \infty),$$

equivalently $f_2(d) = d^{(1+o(1))2^{d-1}}$.

Proof. Lower bound. Let $d \geq 3$ and $m = d - 2$. By Lemma 2.1 and (2),

$$\begin{aligned} \log f_2(d) &\geq 4(\log f_{0,2}(d-2) - 2^{d-2} \log 2) \\ &\geq 4(1 - o(1)) 2^{d-3} \log(d-2) - 2^d \log 2 \\ &= (1 - o(1)) 2^{d-1} \log d, \end{aligned}$$

where the last equality uses $\log(d-2)/\log d \rightarrow 1$ and the fact that the pigeonhole loss $2^d \log 2$ is $O(2^d)$, whereas $2^{d-1} \log d$ is of order $2^d \log d$. Hence

$$\liminf_{d \rightarrow \infty} \frac{\log f_2(d)}{2^{d-1} \log d} \geq 1. \quad (4)$$

Upper bound. By (1), $\log f_2(d) \leq 2^{d-1} \log(d+1) = (1 + o(1)) 2^{d-1} \log d$, so

$$\limsup_{d \rightarrow \infty} \frac{\log f_2(d)}{2^{d-1} \log d} \leq 1. \quad (5)$$

(Remark 3.1 gives an independent proof of (5).)

Together, (4) and (5) give $\log f_2(d) = (1 + o(1)) 2^{d-1} \log d$, i.e. $f_2(d) = d^{(1+o(1))2^{d-1}}$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-03) on the *original* writeup. Its step numbers (S1–S9) refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2401.00299__02/`. Treat it as machine output, not as an authority.

B.1 Interpretation

Problem as posed. Problem 2dim of arXiv:2401.00299 (Problem 1.11 in v1/v2, Problem 1.9 in the catalog’s numbering of the published version) reads: “Determine the asymptotic behaviour of the number $f_2(d)$ ”, preceded by: “It is natural to ask what happens if only 2-dimensional subcubes are allowed to be in a partition. We believe that the same bound as in Proposition [zero2] holds.” Proposition zero2 (v2 Proposition 1.10; folded into Proposition 1.7 with $r=2$ in v3) states $f_{\{0,2\}}(d) = N^{1+o(1)}$, where $n := 2^{\{d-1\}}$ and $N := (d/e)^n$. So the authors’ explicitly conjectured asymptotic is

$$N^{1-o(1)} \leq f_2(d) \leq N^{1+o(1)}, \quad \text{i.e.} \quad \log_2 f_2(d) = (1+o(1)) n \log_2 d.$$

Writeup’s formalization. The writeup proves $\log f_2(d) = (1+o(1)) 2^{\{d-1\}} \log d$, equivalently $f_2(d) = d^{(1+o(1))2^{\{d-1\}}}$. Since $N^{1\pm o(1)} = 2^{\{(1\pm o(1)) n (\log_2 d - \log_2 e)\}}$ and $(\log_2 e)/\log_2 d \rightarrow 0$, the statement “ $f_2(d) = d^{\{(1\pm o(1))n\}}$ ” is *identical* to “ $f_2(d) = N^{1\pm o(1)}$ ” — the constant-per-vertex factor is absorbed by the $o(1)$ in the exponent in both directions. So the writeup proves exactly the bound the authors said they believe, at exactly the precision at which Proposition zero2 itself is stated. No interpretation gaming: the writeup does not weaken the target, and it explicitly discloses that second-order ($2^{\Theta(n)}$) factors are not determined.

Two definitional points I checked against the paper: (i) the paper counts unordered partitions whose classes “span a subhypercube”; for dimension 2 the vertex sets inducing a Q_2 (an induced C_4) in Q_d are precisely the coordinate 2-faces, so the writeup’s “coordinate 2-dimensional subcubes” is the same object; (ii) $f_2 = f_{\{2\}}$ and $f_{\{0,2\}} = f_{\{0,2\}}$ per the paper’s f_S notation — matches the writeup’s usage.

Would the original authors consider it resolved? At the precision they conjectured, yes: the writeup establishes the stated belief. The literal phrase “determine the asymptotic behaviour” is open-ended; a commented-out line in the paper’s TeX (“and that even proving $2^{-(1/2+o(1))n}$ N may be possible, if true”) shows the authors were also curious about the finer exponential-in- n second-order term, which remains open and which the writeup honestly disclaims. This affects scope, not correctness.

B.2 Step-by-step findings

step	label	note
S1. Statement: boxed claim is the conjectured asymptotic (“same bound as Prop. zero2 with $r=2$ ”)	VALID	Equivalence $d^{\{(1\pm o(1))n\}} = N^{1\pm o(1)}$ re-derived ($\log_2 e$ is lower order than $\log_2 d$).

step	label	note
S2. Eq. (1): “Prop. 1.10 gives $\log f_{\{0,2\}}(m) \geq (1-o(1)) 2^{\{m-1\}} \log m$ ”	VALID	Matches the paper: proof of Prop. zero2 gives $f_{\{0,2\}}(m) \geq C(m,2)^{\{(1/2-\delta)2^{\{m-1\}}\}}$ for every fixed $\delta > 0$; $\log_2$ of this is $(1-2\delta)2^{\{m-1\}}(\log_2 m - O(1)) = (1-o(1))2^{\{m-1\}} \log_2 m$. Citation numbering matches arXiv v1/v2 (in v3 it is Prop. 1.7 with $r=2$ — same content; see Reference check).
S3. Pigeonhole (3): some L has $a_L \geq f_{\{0,2\}}(m)/2^{\{2\}m}$	VALID	At most $2^{\{2\}m}$ subsets L of Q_m ; sum of a_L over L equals $f_{\{0,2\}}(m)$; $\max \geq$ mean. Trivially correct.
S4. Construction produces a partition of $Q_{\{m+2\}}$ into 2-subcubes	VALID	$C \times \{z\}$ is a coordinate square (free coords in $[m]$); $\{x\} \times Q_2$ is a coordinate square (free coords $m+1, m+2$). Each layer $Q_m \times \{z\}$ minus $L \times \{z\}$ is exactly covered by the squares of P_z (P_z 's parts are squares plus the singletons L), and $L \times Q_2$ is exactly covered by the verticals. Machine-verified exhaustively for $m=2,3$ and for all 810,000 tuples at $m=4$.
S5. Injectivity in the 4-tuple, hence $f_2(m+2) \geq a_L^4$, giving (2)	VALID	Decoding is deterministic: verticals are exactly the parts with free coordinates $\{m+1, m+2\}$, recovering L ; the layer- z horizontal squares recover P_z 's square parts, and $P_z =$ squares + singletons on L . Distinct tuples give distinct partitions. Machine-verified: $30^4 = 810,000$ distinct images at $m=4$, $3^4 = 81$ at $m=3$.
S6. Lower bound: $\log f_2(d) \geq 4(\log f_{\{0,2\}}(d-2) - 2^{\{d-2\}} \log 2) = (1-o(1)) 2^{\{d-1\}} \log d$	VALID	Re-derived: $4 * (1-o(1)) 2^{\{d-3\}} \log(d-2) = (1-o(1)) 2^{\{d-1\}} \log d$ since $\log(d-2)/\log d \rightarrow 1$; the pigeonhole loss $4 * 2^{\{d-2\}} \log 2 = 2^d \log 2 = o(2^{\{d-1\}} \log d)$. Needs $d \geq 3$ only (asymptotic statement). Gives $\liminf \geq 1$ in (4).

step	label	note
S7. Upper bound (5) via Prop. 1.6: $f_2(d) \leq (d+1)^{\lfloor 2 \log_2 d \rfloor}$	VALID	Paper's Prop. main states $f(d) \leq (d+1)^n$ and $f_2(d) \leq f(d)$ by definition; $\log_2 = n$ $\log_2(d+1) = (1+o(1)) 2^{\log_2 d}$ $\log_2 d$.
S8. "For completeness" elementary upper bound via even-vertex encoding	VALID	Each square of Q_d contains exactly 2 even vertices (antipodal in the square), so "smaller even vertex + free coordinate pair" is an injective code; $C(n, n/2) C(d, 2)^{\lfloor n/2 \rfloor} \leq 2^n d^n = \exp(n \log d + O(n))$. A square partition has $2^{d/4} = n/2$ squares — correct. Redundant to S7 but sound.
S9. Combination of (4) and (5) yields the boxed asymptotic	VALID	Immediate.

No GAP or ERROR found. Every inequality was re-derived independently; the finite construction was verified by code.

B.3 Reference check

- **Source paper obtained directly:** I downloaded the TeX sources of all three arXiv versions of 2401.00299 (v1, v2, v3) and read the statements and proofs verbatim (scratchpad copies; the checks below quote the actual TeX).
- **"Proposition 1.10" (lower bound for $f_{\{0,2\}}$):** In v1/v2 the environment `zero2` is numbered **1.10** and states $f_{\{0,2\}}(d) = N^{1+o(1)}$ (v2; v1 has the typo $N^{1-o(1)}$), with proof $f_{\{0,2\}}(d) \geq \binom{d,2}{1/2-o(1)}^n = N^{1-o(1)}$ via a counting nibble theorem. In v3 this is subsumed into Proposition 1.7 (new), now stated for every fixed $r \geq 2$: $N^{r/2^{r-1}-o(1)} \leq f_{\{0,r\}}(d) \leq N^{r/2^{r-1}+o(1)}$; $r=2$ gives the identical bound. **The writeup's quoted content is exactly what the paper proves;** the numbering "1.10" matches v1/v2 (and the catalog page itself cites "Proposition 1.10" for this bound), while v3 rennumbers it. Confirmed.
- **"Proposition 1.6" (upper bound):** Proposition `main` is numbered 1.6 in v2 and v3 and contains $f(d) \leq (d+1)^n$ (with $n = 2^{\log_2 d}$); since a 2-face partition is a subcube partition, $f_2(d) \leq (d+1)^{\lfloor 2 \log_2 d \rfloor}$ as used. Confirmed. (In v1 this was Theorem 1.6 with the weaker $f(d) \leq (5e)^n N$ — still sufficient for the same asymptotic.)
- **One level deeper (not cited by the writeup but load-bearing for Prop. 1.10):** the paper's proof uses a Frankl–Rödl-nibble counting theorem attributed to Asratian–Kuzjurin; that paper exists: "On the number of nearly perfect matchings in almost regular uniform hypergraphs", Discrete Mathematics 207 (1999), 1–8. The source paper is refereed (Illinois J. Math. 69(1), 109–122, 2025). I did not re-verify the nibble theorem's proof; per the catalog framing, the source paper's propositions are taken as ground truth.
- **Already-known check:** Semantic Scholar citations of 2401.00299 contain no paper resolving $f_2(d)$; the only related follow-up is arXiv:2411.04479 (Tarannikov/Potapov line: partitions of Z_q^n into subcubes of dimension $n-m$, fixed m — a different regime). The bent-function paper arXiv:2108.00232 counts partitions of F_2^m into 2-dimensional **affine subspaces** ($\log_2$ count $\sim (m/2)2^m$, a strictly larger family and much larger count), so it does not resolve f_2 either. I found no publication of the writeup's result; `ALREADY_KNOWN` does not apply.

B.4 Computational check

Scripts in `verification/scripts/2401.00299__02/` (`check_lifting.py`, ~0.06 s; `check_lifting_m4.py`, ~33 s; python3, stdlib only). Exact numbers obtained:

- **Censuses (exact cover, exhaustive):** $f_{\{0,2\}}(1)=1$, $f_{\{0,2\}}(2)=2$, $f_{\{0,2\}}(3)=10$, $f_{\{0,2\}}(4)=339$; $f_2(2)=1$, $f_2(3)=3$, $f_2(4)=30$, $f_2(5)=4380$. Every enumerated 2-face partition of Q_d has exactly $2^{d/4}$ parts, and $f_2(d) \leq (d+1)^{\lfloor d/2 \rfloor}$ holds at every computed value (e.g. $4380 \leq 5^{16}$).
- **Construction validity + injectivity:** for $m=3$, best L = empty set with $a_L = 3$; all $3^4 = 81$ lifted objects are valid coordinate-square partitions of Q_5 , pairwise distinct, and all appear in the exhaustive $f_2(5)$ census ($81 \leq 4380$). For $m=4$, best L = empty set with $a_L = 30$; all $30^4 = 810,000$ lifted objects are valid square partitions of Q_6 and pairwise distinct (so $f_2(6) \geq 810,000$). For $m=2$ ($a_L = 1$) the single image is valid and lies in the $f_2(4)$ census.
- **Lemma (2) numerically:** $f_2(m+2) \geq (f_{\{0,2\}}(m)/2^{\lfloor m/2 \rfloor})^4$ holds for $m = 1, 2, 3$ (right-hand sides 0.0039, 0.00024, $2.3e-06$ — vacuously small at these sizes, as expected: the pigeonhole cost only becomes negligible asymptotically).
- The distribution of singleton sets was also computed (e.g. $m=4$: 182 distinct L out of at most 2^{16}), directly confirming the pigeonhole setup.

No claimed property failed. Everything the construction asserts is machine-confirmed on every case small enough to enumerate.

B.5 Caveats

1. **Precision of “asymptotic behaviour”:** the writeup determines the first-order asymptotic of $\log f_2(d)$ (equivalently $f_2(d) = N^{1 \pm o(1)}$), which is precisely the bound the authors conjecture in the sentence introducing the Problem. It does not determine second-order $2^{\Theta(n)}$ factors (cf. the commented-out remark in the paper’s TeX about $2^{-(1/2+o(1))n} N$); under a maximalist reading of “determine the asymptotic behaviour” that finer question remains open. The writeup states this limitation itself.
2. **Conditionality:** the lower bound inherits the source paper’s Proposition 1.10 (nibble-based, via Asratian–Kuzjurin 1999); the upper bound inherits Proposition 1.6. Both are in a refereed journal paper and were verified verbatim against the TeX, but were not re-proved from scratch here (the writeup also supplies its own valid elementary upper bound, so only the lower bound is genuinely external).
3. **Citation numbering is version-dependent:** “Proposition 1.10” is correct for arXiv v1/v2 (and the catalog page); in v3 the same statement is Proposition 1.7 with $r=2$. A published version of the writeup should cite by content or by the journal version.
4. **Small- d edge cases:** the lifting needs $m = d-2 \geq 1$ and the bounds are asymptotic; inequality (2) is vacuous ($\text{RHS} < 1$) for small m , which is harmless. $f_2(d) \geq 1$ for all $d \geq 2$, so the $\liminf$ statement is well-posed.
5. **Convention:** logs must be read in a fixed base throughout (the paper uses base 2); the chain in the “Lower bound” section is base-independent as written. The identification $Q_{\{m+2\}} = Q_m \times Q_2$ fixes the last two coordinates as the Q_2 factor; the decoding argument depends on this and is used consistently.
6. **Sociological note, not a finding:** the argument is strikingly simple for a problem left open by Alon, Balogh and Potapov; the most plausible explanation is that the authors regarded the finer (second-order) asymptotics as the real content of “determine the asymptotic behaviour”. I searched for, and failed to find, any mathematical flaw, and found no prior publication of the observation.

B.6 Referee summary

The writeup's single new ingredient is a lifting lemma: pigeonhole the $\{0,2\}$ -partitions of Q_{d-2} by their singleton set L (cost at most $2^{\lfloor d-2 \rfloor}$ per layer, i.e. $2^d \log 2$ in total after taking fourth powers — asymptotically negligible against $2^{d-1} \log d$), pick four partitions with the same L , lay them on the four layers of $Q_{d-2} \times Q_2$, and replace the four aligned copies of each singleton by one vertical square. Every step of this argument is correct as written; the decoding argument makes the count injective, so $f_2(d) \geq (f_{\{0,2\}}(d-2)/2^{\lfloor d-2 \rfloor})^4$, and the source paper's Proposition 1.10 ($f_{\{0,2\}}(m) \geq N_m^{1-o(1)}$, verified verbatim in the paper's TeX) then gives $\log f_2(d) \geq (1-o(1)) 2^{d-1} \log d$. The matching upper bound is the paper's Proposition 1.6 (also verified; the writeup's independent encoding bound is likewise valid). I re-derived every inequality, confirmed both cited propositions against all three arXiv versions, machine-verified the construction exhaustively (all 810,000 lifted 4-tuples at $m=4$ give distinct valid square partitions of Q_6 ; exhaustive censuses $f_2(2..5) = 1, 3, 30, 4380$ agree with all claims), and found no resolving prior work. The result proves exactly the asymptotic the authors conjectured ($f_2(d) = N^{1 \pm o(1)}$); only finer second-order asymptotics remain open, as the writeup itself states. CONFIRMED.

References

- [1] N. Alon, J. Balogh and V. N. Potapov, Partitioning the hypercube into smaller hypercubes, *Illinois J. Math.* 69 (2025) 109–122; arXiv:2401.00299.
- [2] A. S. Asratian and N. N. Kuzjurin, On the number of nearly perfect matchings in almost regular uniform hypergraphs, *Discrete Math.* 207 (1999) 1–8.